

Easton Yi HUANG

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EDUCATION

Ph.D. in Mechanical Engineering

Rutgers University–New Brunswick, School of Engineering

Aug. 2024 - Present

M.Sc. in Mechanical Engineering

National University of Singapore, College of Design and Engineering

Aug. 2022 - Jan. 2024

B.Eng. in Mechanical Design & Manufacturing and Their Automation

Dalian University of Technology, School of Mechanical Engineering

Sep. 2018 - Jul. 2021

NUS Suzhou Research Institute

Sep. 2021 - Jul. 2022

RESEARCH EXPERIENCE

NextG-Enabled Manufacturing of the Future (NextGEM)

Rutgers, New Brunswick, USA

Instructor: Prof. Jingang Yi & Prof. Yuebin Guo

2024 - Present

- Keywords: Physics-Informed Machine Learning; Digital Twins; Manufacturing.

Digital Twins Integrated Finite Element Analysis

NUS, Singapore

Instructor: Prof. Andrew Yeh Ching Nee & Prof. Ong Soh Khim

2022 - 2024

- Keywords: Digital Twins; Structural Health Monitoring; Finite Element Analysis; Machine Learning; Surrogate Model.

Intelligent Machine Vision for Surface Condition Inspection

NUSRI, Soochow, China

Instructor: Prof. Wen Feng Lu

2021 - 2022

- Keywords: Machine Vision; Defect Detection; Deep Learning; Denoising.

OTHER EXPERIENCE

- **Teaching Assistant, Course: 14:440:291–Honors Engineering Mechanics: Statics (Spring 2026)**

2025-2026, Rutgers University, Advisor: Prof. Onur Bilgen

- **Teaching Assistant, Course: 14:650:401–Dynamic Systems and Controls (Fall 2025)**

2025-2026, Rutgers University, Advisor: Prof. Annalisa Scacchioli

- **Graduate Assistant**

2024-2025, Rutgers University, Advisor: Prof. Jingang Yi

- **China Robotics and Artificial Intelligence Competition**

Smart Agriculture Contest, Instructors: Dr. Feilong Wang & Prof. Shenglan Liu

- **Kaggle Competition**

UW-Madison GI Tract Image Segmentation (UWMGI)

- **Dalian University of Technology Varsity Self-Reliance Society**

Vice President (Junior), Office Manager (Sophomore), Outreach Officer (Freshman)

AWARDS

- Raisler Fellowship from Rutgers University.
- Dongguan Entrepreneur Scholarship in 2024.
- First Prize in China Robotics and Artificial Intelligence Competition.
- Learning Excellence Award in DUT.
- Honorary title of Outstanding Officer of Dalian University of Technology Self-Reliance Society.

PUBLICATIONS

- J1. **Y. Huang**, J. Yi, Y. Guo. “Physics-Driven Real-Time, Intelligent Digital Twin for Adaptive Milling Chatter Control,” *CIRP Ann. - Manuf. Technol.*, 2026, Turin, Italy.
- C1. **Y. Huang**, F. Han, J. Yi, Y. Guo. “Milling iTwin: An Intelligent Digital Twin of Milling Dynamics for Process Monitoring and Chatter Suppressions,” in *Proc. IEEE Int. Conf. Autom. Sci. Eng. (CASE)*, 2026, Shenyang, China. (Accepted)
- C2. **Y. Huang**, F. Han, W. Liu, J. Yi, Y. Guo. “Machine Learning-based Online Stability Lobe Diagram Estimation and Chatter Suppression Control in Milling Process,” in *Proc. Amer. Control Conf. (ACC)*, 2026, New Orleans, LA, USA. (Accepted)
- C3. F. Han, **Y. Huang**, J. Yi. “Active Training Data Selection for Gaussian Process-Based Robot Dynamics Learning and Control,” in *Proc. IEEE/RSJ Int. Conf. on Intell. Robots and Syst. (IROS)*, 2025, Hangzhou, China.
- C4. **Y. Huang**, F. Han, J. Yi. “Data-Efficient Learning-Based Estimation of Region of Attractions for Nonlinear Dynamic Systems,” in *Proc. IEEE Int. Conf. Autom. Sci. Eng. (CASE)*, 2025, Los Angeles, CA, USA.
- C5. **Y. Huang**, F. Han, T. Zheng, L. Hu, J. Yi, Y. Guo. “Physics-Informed Machine Learning-Based Chattering Prediction in Milling Process,” in *Proc. Amer. Control Conf. (ACC)*, 2025, Denver, CO, USA.
- C6. **Y. Huang**. “Intelligent Machine Vision for Detection of Steel Surface Defects with Deep Learning,” in *Proc. IEEE Int. Conf. Smart Internet Things (SmartIoT)*, 2023, Xining, China.
- W1. **Y. Huang**, J. Yi, Y. Guo. “A Finite Element-Driven Intelligent Digital Twin of Milling Dynamics for Real-Time Monitoring and Chatter Suppression,” *ICRA Workshop on Advances and Challenges in AI-Driven Automation and Robotic System Integration with Digital Twins*, 2026, Vienna, Austria.

PAPER REVIEW

- IEEE Transactions on Control Systems Technology 2026
- International Journal of Precision Engineering and Manufacturing 2026
- IEEE International Conference on Advanced Intelligent Mechatronics 2026
- American Control Conference 2025, 2026
- IEEE International Conference on Automation Science and Engineering 2025, 2026
- IEEE International Conference on Intelligent Transportation Systems 2025
- IEEE International Conference on Robotics and Automation 2026